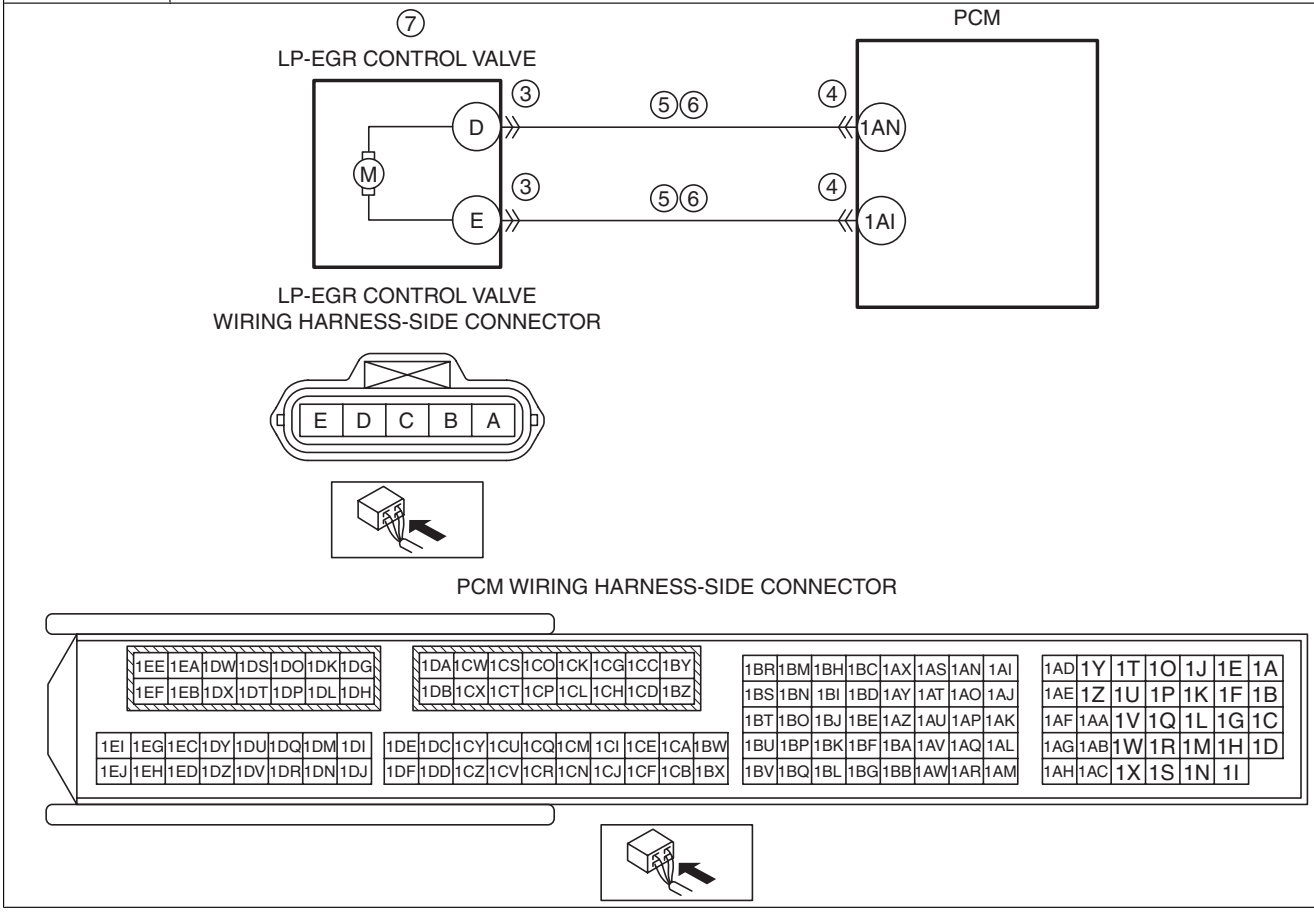
DTC P213B:00 [SKYACTIV-D 1.5]

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DTC P213B:00	Circuit malfunction in LP-EGR control valve DC motor control system
DETECTION CONDITION	• When the following conditions are met, if the LP-EGR control valve duty value is **90 % or more** for **2 s**, the PCM determines that there is a LP-EGR control valve control system range/performance problem. **MONITORING CONDITIONS** — LP-EGR control: during feedback control — The following DTCs are not detected: • LP-EGR EGR control valve DC motor control system: P045B:00 • LP-EGR control valve position sensor: P0407:00, P0408:00 **Diagnostic support note** • This is a continuous monitor (EGR system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• LP-EGR control valve connector or terminals malfunction • PCM connector or terminals malfunction • Short to power supply in wiring harness between the following terminals: — LP-EGR control valve terminal D—PCM terminal 1AN — LP-EGR control valve terminal E—PCM terminal 1AI • Short to ground in wiring harness between the following terminals: — LP-EGR control valve terminal D—PCM terminal 1AN — LP-EGR control valve terminal E—PCM terminal 1AI • LP-EGR control valve malfunction • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED REPAIR INFORMATION AVAILABILITY** • Verify related Service Bulletins and/or on-line repair information availability. • Is any related repair information available?	Yes	Perform repair or diagnosis according to the available repair information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT LP-EGR CONTROL VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the LP-EGR control valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.
4	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.
5	**INSPECT LP-EGR CONTROL VALVE CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the LP-EGR control valve and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the following terminals (wiring harness-side): — LP-EGR control valve terminal D — LP-EGR control valve terminal E • Is the voltage **0 V**?	Yes	Go to the next step.
		No	**If there is no common connector:** Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • LP-EGR control valve terminal D—PCM terminal 1AN • LP-EGR control valve terminal E—PCM terminal 1AI **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. • Repair or replace the wiring harness which has a short to power supply. Go to Step 8.
6	**INSPECT LP-EGR CONTROL VALVE CIRCUIT FOR SHORT TO GROUND** • Verify that the LP-EGR controlvalve and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — LP-EGR control valve terminal D — LP-EGR control valve terminal E • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • LP-EGR control valve terminal D—PCM terminal 1AN • LP-EGR control valve terminal E—PCM terminal 1AI **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 8.
		No	Go to the next step.
7	**INSPECT LP-EGR CONTROL VALVE** • Inspect the LP-EGR control valve. (See LP-EGR CONTROL VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction**?**	Yes	Replace the LP-EGR control valve, then go to the next step. (See LP-EGR CONTROL VALVE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
8	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Perform the KOEO self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
9	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.